

akt Console CLI — End-to-End Test Report

Subject: akash-network/akt v0.0.4 (commit 8dd6195, released 2026-08-05)

Date: 2026-08-12 | **Binary:** akt_0.0.4_darwin_all.zip (darwin/arm64)

Environment: isolated AKT_HOME, mainnet Console API, funded production account

Issues filed: <https://github.com/akash-network/akt/issues/57> through #67

This report documents thorough end-to-end testing of the **Akash Console integration** in the unified akt CLI, plus a lighter smoke pass over non-Console surfaces (chain query/tx help, keyring, store, monitor/MCP help). GitHub issues were filed for every actionable finding.

Coverage at a glance

Surface	Depth	Result
akt console * (auth, wallet, deploy lifecycle, gateway, marketplace, keys/JWT)	Thorough E2E	Functional PASS; UX gaps
akt sdl (scaffolds / init / validate)	Thorough	PASS — excellent
akt deploy / update / close (console-api rail)	Thorough E2E	PASS with P0 scripting gap
context / network / action log / store	Smoke + setup	PASS (bootstrap noise)
query / tx (chain)	Smoke only	PARTIAL — RPC works; no keyring deploys
keys / keyring (test backend)	Smoke	PASS (add/list)
monitor TUI / MCP live / provider mTLS	Help-only	SKIP interactive

Did we test non-Console commands thoroughly? No. Non-Console was smoke-tested only: help text, a few live chain queries against mainnet RPC, key add/list on an isolated test keyring, store status, and workflow dry-runs. We did **not** run full keyring-signed deployments, monitor TUI sessions, MCP client sessions, provider mTLS flows, or Cosmos module tx broadcast suites. Those remain follow-up work.

1. Executive summary

The Console rail is **functionally solid for the full lifecycle**: login → SDL → screen → deploy → status/logs/events/shell → update → deposit → settings → close. Auth, capability gating, and the action log work as designed. SDL offline tooling is best-in-class.

Biggest gaps: **agent/scripting UX** (JSONL missing dseq; pretty mode dumps raw API JSON) and **parity with console-axi** (no deep links, no URI wait, no resource-flag screening, weaker money formatting).

Severity	Issues	Themes
P0 – fix soon	#57 #58 #59	JSONL no dseq; deposit min ignored; raw API 400 blobs
P1 – important	#60–#65	Pretty tables, URIs/links, screen flags, bootstrap noise, shell UX, money fmt
P2 – polish	#66 #67	Agent help[], list filters, query owner default on console-api

2. Environment & method

CLI: akt 0.0.4, go1.26.2, darwin/arm64 universal binary from GitHub Releases.

Home: AKT_HOME=/tmp/akt-e2e-home (isolated; not ~/.config/akt).

Auth: Console API key via akt console login + env override.

Network: mainnet template (akashnet-2), RPC https://rpc.akt.dev.

Mutating tests: real deploys with \$0.50 deposits; session e2e deployments closed; temp API key deleted.

Transcripts: /tmp/akt-e2e-results/.

3. Command matrix — Console & workflows (thorough)

Command / flow	Result	Notes
console login / logout / whoami	PASS	Key 0600; gating works
wallet balance / list / settings / cost	PASS	USD formatting good
usage [from] [to]	PASS	Bad dates fail cleanly
deployment list / get	PASS	Raw JSON pretty; pagination ok
deployment create / update / close	PASS	Min \$0.50 on create
deployment deposit	FAIL	Accepts \$0.01 (no min) — #58
deployment settings true/false	PASS	Auto top-up toggle
bid list + lease create	PASS	Atomic path works
logs / events / status	PASS	JWT gateway path
shell <dseq> <svc> -- cmd	PARTIAL	Needs service name — #64
screen <sdl>	PASS	No resource flags — #62
provider list/get/regions/auditors	PASS	Default limit 20
gpu / template list/get/sdl	PASS	Template SDL raw YAML
apikey create/list/delete	PASS	Positional name only
jwt create --ttl --scope	PASS	
sdl scaffolds/init/validate	PASS	Rejects :latest; great hints
akt deploy (pretty)	PASS	Prints DSEQ; no URIs — #61
akt deploy -o jsonl	FAIL	No dseq in output — #57
akt deploy --dry-run	PARTIAL	Skips min deposit check — #58
capability gating (no key)	PASS	[unavailable] + remedy
action log	PASS	Records dseq mutations

4. Command matrix — Non-Console (smoke only)

Light pass to establish what works without claiming full coverage.

Command / flow	Result	Notes
version / version --long	PASS	Build metadata correct
completion bash	PASS	Script emitted
context list/show/network/log	PASS	Pretty tables good
context keys add/list (test backend)	SMOKE	Isolated home only
query block (live RPC)	SMOKE	Mainnet height returned
query staking params	SMOKE	Pretty params table
query provider --limit 1	SMOKE	Provider table
query deployment / market bid	PARTIAL	Needs default-account — #67
query bank params	PARTIAL	Subcommand name differs
tx help / bank help	SMOKE	Help only; no broadcast
provider --help	SMOKE	Gateway group present
monitor --help	SKIP	Interactive TUI not launched
mcp --help	SMOKE	stdio MCP help
store status	PASS	Empty store on console ctx
store list	FAIL	Command does not exist
close / update --dry-run	SMOKE	Workflow plan prints
Full keyring deploy / cert / manifest	SKIP	Not exercised
Monitor dashboards live	SKIP	Not exercised
IBC / WASM / gov txs	SKIP	Not exercised

Recommended follow-up non-Console E2E: (1) keyring context with funded/sandbox account; (2) akt deploy on chain rail with cert + manifest; (3) monitor dashboards for N minutes; (4) MCP tools/list with and without --enable-writes; (5) provider lease status with JWT; (6) representative tx dry-runs per module group.

5. What works well

- **Full Console deploy lifecycle is fast and reliable** (~10–25s). Manifest step correctly skipped on Console rail.
- **SDL offline tooling is excellent:** rejects :latest and bad denoms; clear fix hints; four scaffolds.
- **Capability gating** marks [unavailable] and fails fast with login remedy.
- **Action log** records console mutations with dseq — safety net when stdout is incomplete.
- **Gateway ops** (logs, events, status, shell) work via Console-minted JWTs without local keys.
- **Security hygiene:** API key file mode 0600; not in config.yaml; apikey create warns “shown once”.
- **Idempotent close**, deposit formats (0.5 / 0.5usd / \$0.50), settings toggle all work.
- **Pretty akt deploy** prints a clear DSEQ line.

6. Findings & filed GitHub issues

P0 — #57: deploy -o jsonl omits dseq and step outputs (unscriptable)

JSONL steps complete with empty useful payload. dseq only in action log. Blocks CI/agents.

<https://github.com/akash-network/akt/issues/57>

P0 — #58: console deployment deposit accepts amounts below \$0.50 minimum

Create/deploy enforce MinConsoleDepositUSD; deposit only checks amount > 0. Dry-run also skips min.

<https://github.com/akash-network/akt/issues/58>

P0 — #59: console gateway/get leak raw API 400 JSON for non-numeric dseq

logs/get/status with 'notanumber' returns Zod validation blob. Need client-side uint64 parse.

<https://github.com/akash-network/akt/issues/59>

P1 — #60: console pretty output dumps raw API JSON (no human/table view)

pretty ~ json for list/get/status/bids. Need tables + summaries; keep -o json for wire shape.

<https://github.com/akash-network/akt/issues/60>

P1 — #61: deploy success should print URIs, Console deep-link, auto-top-up opt-out

No URIs, no console.akash.network link, autoTopUpEnabled true without printed disable command.

<https://github.com/akash-network/akt/issues/61>

P1 — #62: console screen should accept resource flags (not only SDL file)

Parity with console-axi: --cpu/--memory/--gpu and SDL overrides.

<https://github.com/akash-network/akt/issues/62>

P1 — #63: bootstrap wizard warning printed on successful network/context create

Empty home non-TTY: scary wizard message then success exit 0.

<https://github.com/akash-network/akt/issues/63>

P1 — #64: console shell: require service name clearly; default when only one service

Missing service → 'kube: no service for that lease'. Should list services or default.

<https://github.com/akash-network/akt/issues/64>

P1 — #65: console money display: format uact escrow/lease prices as USD in pretty mode

Wallet uses \$; escrow 500000 uact after \$0.50 deposit; lease prices opaque.

<https://github.com/akash-network/akt/issues/65>

P2 — #66: console agent UX parity: next-step help, list filters, quieter create success

Bundled polish: help[], state filters, context create ack, apikey --name, empty events, store list.

<https://github.com/akash-network/akt/issues/66>

P2 — #67: query deployment/market require default-account even for console-api contexts

Managed wallet address not used as query owner default; friction for console-only users.

<https://github.com/akash-network/akt/issues/67>

7. Detailed flow notes

7.1 Auth & context

console-api context + login works. Env AKT_CONSOLE_API_KEY resolves before stored key. Logout clears stored credential and gates authenticated commands. context show reports Capabilities correctly.

7.2 SDL

Default scaffold image nginx:1.27, pricing denom uact. :latest rejected at init (exit 2). Pipe validate works. Unpinned image + uakt denom produce structured errors/warnings with hints.

7.3 Deploy workflow (console-api)

```
Deployment active!
DSEQ: 1786522800616
Results:
create-deployment success tx: DFF26856...
wait-for-bids / select-bid / create-lease / display-result success
Workflow "deploy" completed successfully.
```

JSONL mode for the same flow omitted dseq entirely (#57). send-manifest skipped with a stderr note (correct). Dry-run still shows chain message type names. Low deposit fails at create with a clear error on real run, but dry-run accepts \$0.01.

7.4 Atomic path

create → bid list → lease create → status → close all worked. Create returns clean {dseq, txHash, state}. Lease create returns a full get-shaped deployment blob.

7.5 Gateway

Logs with --tail and optional service work. Events good on fresh deploy. Status after close: clear error, exit 1. Shell exec works with explicit service name.

7.6 akt vs console-axi

Capability	akt 0.0.4	console-axi 1.1.1
Output	Pretty ≈ JSON	Compact TOON
Next-step help	No	help[] everywhere
Deploy URIs	No	Yes
Console deep links	Not observed	Yes
Screen resource flags	No	Yes
Monthly cost on list	No	Yes
Deposit min on top-up	No (\$0.01 ok)	Client min \$0.50
Unified chain+console	Yes	Console-only
Action log / multi-context	Yes	Minimal

Takeaway: akt is the right long-term surface. Closing JSONL/dseq/URI/pretty gaps makes it the better default.

8. Recommended priority order

1. Emit dseq (+ provider, URIs) in deploy JSONL and pretty summary — #57, #61
2. Enforce or document deposit minimum consistently — #58
3. Pretty tables for deployment list/get/status/bids with USD formatting — #60, #65
4. Post-deploy: wait for ready + URIs + console link + auto-top-up opt-out — #61
5. Screen resource flags — #62
6. Client-side dseq validation + cleaner API errors — #59
7. Shell service default / better errors — #64
8. Silence bootstrap warning on network/context create — #63

9. Agent help[] + list filters + misc polish — #66

10. Console-api owner default for chain queries — #67

9. Suggested regression checks

```
# 1) JSONL must expose dseq
akt deploy app.yaml --deposit 0.5 --bid-select cheapest --yes -o jsonl \
| jq -e 'select(.step=="create-deployment") | .outputs.dseq // .dseq'

# 2) Deposit min consistent
akt console deployment deposit <dseq> 0.01 # expect fail if min is policy
akt console deployment create app.yaml 0.01 # expect fail

# 3) Ready + URI
akt deploy app.yaml --deposit 0.5 --bid-select cheapest --yes \
| grep -E 'http|ingress|console.akash'

# 4) Gating
akt console logout && akt console whoami # expect unavailable + exit 1
```

10. Test hygiene

- All intentional e2e deployments created in this session were closed.
- Small test deposits (\$0.01×2, \$0.50× few) hit existing and short-lived deployments.
- Temporary API key akt-e2e-temp-* was created and deleted.
- Pre-existing active deployments on the account (~13) left running except tiny deposit side effects.
- Ephemeral keyring test key only under /tmp/akt-e2e-keyring-home; discard that home.
- Command transcripts: /tmp/akt-e2e-results/.

Report generated 2026-08-12 for akash-network/akt. Issues #57–#67 at <https://github.com/akash-network/akt/issues>